

Two-stage experiments

1. Luther and Kyle are playing a game. Luther flips two coins and Kyle flips three. Kyle wins if he has more heads flips than Luther. What is the probability that Kyle wins?

Follow the two-stage experiment strategy. Let A be the event that Kyle wins. Let B_0 be the event that Luther flips 0 heads, B_1 be the event that Luther flips 1 head, and B_2 be the event that Luther flips 2 heads. The events B_0, B_1, B_2 form a partition in the sense that $B_0 \cup B_1 \cup B_2 = \Omega$ and the events are disjoint.

The law of total probability tells us $P(A) = P(B_0) P(A \mid B_0) + P(B_1) P(A \mid B_1) + P(B_2) P(A \mid B_2)$. Use this formula to calculate $P(A)$ as follows:

Calculate $P(B_0), P(B_1), P(B_2)$. (As a shortcut, use the fact that the number of heads Luther flips is a random variable with the binomial(2, 1/2) distribution.)

$P(B_0) =$

$P(B_1) =$

$P(B_2) =$

Calculate $P(A \mid B_0), P(A \mid B_1), P(A \mid B_2)$.

$P(A \mid B_0) =$ (Probability that Kyle wins given that Luther flips 0 heads. How many heads does Kyle need to flip to win in this circumstance?)

$P(A \mid B_1) =$ (Probability that Kyle wins given that Luther flips 1 head. How many heads does Kyle need to flip to win in this circumstance?)

$P(A \mid B_2) =$ (Probability that Kyle wins given that Luther flips 2 heads. How many heads does Kyle need to flip to win in this circumstance?)

Use the law of total probability to calculate $P(A)$.

$$P(A) =$$

2. Imagine the chance experiment of rolling a (standard 6-sided) die and flipping as many coins as the number you see on the die. What is the probability that you flip exactly 3 heads?

Follow the two-stage experiment strategy. Let A be the event that you flip 3 heads. Let B_i be the event “you roll an i ,” where i runs from 1 to 6. These six events form a partition.

The law of total probability tells us $P(A) = P(B_1) P(A \mid B_1) + P(B_2) P(A \mid B_2) + \dots + P(B_6) P(A \mid B_6)$. Use this formula to calculate $P(A)$.